

## Query optimization:

```
ii = input().split(" ")
n,q = int(ii[0]),int(ii[1])
ii = input().split(" ")
lst = []
for i in range(n):
    lst.append(int(ii[i]))
dic = {}
for x in lst:
    if (not(x in dic.keys())):
        dic[x] = 0;
    dic[x]+=1
for i in range(q):
    ii = input().split(" ")
    if(int(ii[0])==1):
        p = int(ii[1])
        x = int(ii[2])
        y = lst[p]
        dic[y]-=1
        if (not(x in dic.keys())):
            dic[x] = 0;
        dic[x]+=1
        lst[p] = x
    else :
        x = int(ii[1])
        if (not(x in dic.keys())):
            print(0)
        else :
            print(dic[x])
```

## Special numbers :

```
L=[int(i) for i in input().split()]
x=int(input())
```

```

L.sort()
if x in L:
    print(L.index(x))
else:
    L.append(x)
    L.sort()
    print(L.index(x))

```

## Count Pairs :

```

ii = input().split(" ")
n,m = int(ii[0]),int(ii[1])
ii = input().split(" ")
a = []
b = []
for i in range(n):
    a.append(int(ii[i]))
ii = input().split(" ")
for i in range(m):
    b.append(int(ii[i]))
sm = int(input())
dic = {}
for x in a:
    if (not(x in dic.keys())):
        dic[x] = 0;
    dic[x]+=1
ans = 0
for i in range(m):
    tt = sm - b[i]
    if tt in dic.keys():
        ans += dic[tt]
print(ans)

```

## Code am I? :

```
import math
firstList = [int(x) for x in input().split()]
T = int(input())

newList = list()

for element in firstList:
    if element < T:
        newList.append(int(math.pow(element, 2)))
        ##or --> newList.append(element**2)
    else:
        newList.append(element*2)

toDictionary = dict(zip(firstList, newList))
print(toDictionary)
```

## Modify List :

```
import math
one = [int(x) for x in input().split()]

two = list()

n = math.ceil(len(one)/2)

for pos in range(n):
    if pos != len(one)-1-pos :
        two.append(one[pos]+one[len(one)-1-pos])
    else:
        two.append(one[pos])
print(*two)
```

## Zero to centre :

```
n = int(input())
lst = []
x = -1
y = -1;
for i in range(n):
```

```

ll = input().split(" ");
lst.append(ll)
for i in range(n):
    for j in range(n):
        if lst[i][j]=='0':
            x = i;
            y = j;

ans = abs(x - n//2) + abs(y-n//2)
print(ans)

```

## Closest vowel :

```

inp = input()
s = "abcdefghijklmnopqrstuvwxyz"

v = "aeiou"

stor = []
for i in v:
    stor.append(s.index(i))

ans = ""
for i in inp:
    val = s.index(i)
    coun = 10**5
    apl = i
    for j in range(len(stor)):
        if abs(val - stor[j]) < coun:
            coun = abs(val - stor[j])
            apl = v[j]
    ans += apl
print(ans)

```